

Calculus I — Limits and Derivatives

1.1 The Concept of a Limit

A limit is defined as the value that a function $f(x)$ approaches as the input x approaches some particular value, without necessarily ever reaching it. Formally, we write $\lim_{x \rightarrow a} f(x) = L$ to indicate that $f(x)$ gets arbitrarily close to L as x gets arbitrarily close to a .

The formal definition of a limit, often called the epsilon-delta definition, states that $\lim_{x \rightarrow a} f(x) = L$ if for every epsilon greater than zero there exists a delta greater than zero such that whenever $0 < |x - a| < \delta$, it follows that $|f(x) - L| < \epsilon$. This rigorous definition underlies all of calculus.

One-sided limits refer to the behavior of a function as x approaches a from only one direction: the left-hand limit, written $\lim_{x \rightarrow a^-} f(x)$, and the right-hand limit, written $\lim_{x \rightarrow a^+} f(x)$. A two-sided limit exists if and only if both one-sided limits exist and are equal.

1.2 Limit Laws and Techniques

The sum law states that $\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$, provided both limits exist. Similarly, the product law states that $\lim_{x \rightarrow a} [f(x) * g(x)] = \lim_{x \rightarrow a} f(x) * \lim_{x \rightarrow a} g(x)$.

Direct substitution is the simplest technique for evaluating a limit: if the function is continuous at $x = a$, then $\lim_{x \rightarrow a} f(x) = f(a)$. When direct substitution produces an indeterminate form such as $0/0$, algebraic techniques such as factoring, rationalizing the numerator, or applying L'Hopital's Rule are required instead.

L'Hopital's Rule states that if $\lim_{x \rightarrow a} f(x)/g(x)$ produces an indeterminate form of type $0/0$ or infinity/infinity, then $\lim_{x \rightarrow a} f(x)/g(x) = \lim_{x \rightarrow a} f'(x)/g'(x)$, provided the latter limit exists. This rule is frequently tested and frequently misapplied when the indeterminate form condition is not actually met.

1.3 Continuity

A function f is defined as continuous at a point $x = a$ if three conditions are satisfied: $f(a)$ is defined, the limit of $f(x)$ as x approaches a exists, and the limit equals $f(a)$. If any of these three conditions fails, the function is said to be discontinuous at that point.

There are three common types of discontinuity: removable discontinuity, where the limit exists but does not equal the function value or the function value is undefined; jump discontinuity, where the left-hand and right-hand limits exist but are not equal; and infinite discontinuity, where the function approaches positive or negative infinity near the point.

1.4 The Derivative Defined

The derivative of a function f at a point x is defined as $f'(x) = \lim_{h \rightarrow 0} [f(x + h) - f(x)] / h$, provided this

limit exists. Geometrically, the derivative represents the slope of the tangent line to the curve $y = f(x)$ at the point x .

The derivative can also be interpreted as the instantaneous rate of change of a function with respect to its independent variable. This interpretation is what makes derivatives so useful in physics, economics, and engineering, where quantities such as velocity, marginal cost, and growth rate are all derivatives of other functions.

A function is said to be differentiable at a point if its derivative exists at that point. A key theorem states that if a function is differentiable at a point, it must also be continuous at that point, although the converse is not true: a function can be continuous but not differentiable, as in the case of a sharp corner in the graph.

1.5 Differentiation Rules

The power rule states that if $f(x) = x^n$, then $f'(x) = n * x^{(n-1)}$. This is one of the most frequently used differentiation rules and applies to any real exponent n .

The product rule states that if $f(x) = u(x) * v(x)$, then $f'(x) = u'(x) * v(x) + u(x) * v'(x)$. The quotient rule states that if $f(x) = u(x) / v(x)$, then $f'(x) = [u'(x) * v(x) - u(x) * v'(x)] / [v(x)]^2$.

The chain rule states that if $f(x) = g(h(x))$, then $f'(x) = g'(h(x)) * h'(x)$. This rule is essential for differentiating composite functions and is one of the topics students find most difficult, since it requires correctly identifying the outer and inner functions before applying the formula.

1.6 Applications of Derivatives

A critical point of a function f is defined as a value of x in the domain of f where $f'(x) = 0$ or $f'(x)$ does not exist. Critical points are essential for locating local maxima and local minima using the first derivative test or the second derivative test.

The first derivative test states that if $f'(x)$ changes from positive to negative at a critical point, that point is a local maximum; if $f'(x)$ changes from negative to positive, that point is a local minimum. The second derivative test uses the sign of $f''(x)$ at a critical point instead: a positive second derivative indicates a local minimum, while a negative second derivative indicates a local maximum.

Related rates problems use the chain rule to relate the rates of change of two or more related quantities with respect to time. A common exam tip is to always write down what is given, what is being asked for, and the equation relating the variables before differentiating both sides with respect to time.

1.7 Chapter Summary and Review Questions

In summary, a limit describes the value a function approaches near a point, continuity requires that a function have no gaps or jumps, and the derivative measures the instantaneous rate of change of a function, defined formally as a limit of a difference quotient. Mastery of the power rule, product rule, quotient rule, and chain rule is essential for computing derivatives efficiently.

Students preparing for an exam on this chapter should be able to evaluate limits using algebraic techniques and L'Hopital's Rule, determine whether a function is continuous at a given point, compute derivatives using all four major differentiation rules, and apply derivatives to find critical points and solve related rates problems.

Review question: What are the three conditions required for a function to be continuous at a point?

Review question: State the chain rule and explain when it should be applied. Review question: How is the second derivative test used to classify critical points?